Linear Algebra Software

A Small Selection

11/02/2026

Cédric Chevalier

Contents

1 Introduction 3
1.a Disclaimer 4
2 Dense Linear Algebra 5
2.a Algebra Computations 6
2.b Solvers 7
3 Sparse Linear Algebra 8
3.a Direct Solvers 9
3.b Iterative Solvers (1) 10
3.c Iterative Solvers (2) 11
3.d Eigen Values 12
4 Related 13
4.a Partitioners 14
4.b More 15

1 Introduction

Disclaimer



The lists are non exhaustive

2 Dense Linear Algebra

Algebra Computations

- BLAS implementations:
  - Generic: ATLAS, OpenBLAS, BLIS, GotoBLAS, …
  - Vendors: MKL, ARMPL, ESSL, Cray, CuBLAS, HipBLAS, …
- Toolkits:
  - Eigen
  - CUTLASS

Solvers

- LAPACK (Linear Algebra Package)
- ScaLAPACK
- TLAPACK
- PLASMA (Parallel Linear Algebra Software for Multicore Architectures)
- MAGMA (Matrix Algebra on GPU and Multi-core Architectures)
- Chameleon (https://project.inria.fr/chameleon/)

3 Sparse Linear Algebra

Direct Solvers

- MUMPS: MUltifrontal Massively Parallel sparse direct Solver (https://mumps-solver.org/)
- PaStiX: Parallel Sparse matriX package (https://solverstack.gitlabpages.inria.fr/pastix/)
- SuperLU: Supernodal LU (https://portal.nersc.gov/project/sparse/superlu/)
- PARDISO
- SuiteSparse

Iterative Solvers (1)

- hypre (https://computing.llnl.gov/projects/hypre-scalable-linear-solvers-multigrid-methods)
  - AMG
  - Incomplete Factorizations
- Ginkgo (https://ginkgo-project.github.io)
  - Incomplete Factorizations
- PETSc: the Portable, Extensible Toolkit for Scientific Computation (https://petsc.org)
  - MG/AMG
  - Various Preconditioners
- Trilinos (https://trilinos.github.io)
  - AMG (MueLU)
  - Combinatorial Preconditioners
  - Incomplete Factorizations (Ifpack2)
  - Hybrid Solvers (ShyLU)

Iterative Solvers (2)



- AmgX (https://developer.nvidia.com/amgx)
- rocALUTION (https://rocm.docs.amd.com/projects/rocALUTION)
  - Incomplete Factorizations
  - Approximate Inverse

Eigen Values

- SLEPc: Scalable Library for Eigenvalue Problem Computations (https://slepc.upv.es)
- Trilinos (Anazasi)
- ARPACK: the ARnoldi PACKage (https://www.arpack.org)

4 Related

Partitioners

- METIS / ParMETIS: Parallel Graph Partitioning and Fill-reducing Matrix Ordering (https://github.com/KarypisLab/ParMETIS)
- Scotch (https://www.labri.fr/perso/pelegrin/scotch)
- Zoltan (https://www.sandia.gov/ccr/software/zoltan/), part of Trilinos
- KaHyPar: Karlsruhe Hypergraph Partitioning (https://kahypar.org)
- PaToH
- Mondriaan

More

- https://www.netlib.org/utk/people/JackDongarra/la-sw.html